Topic 4: Swift Collections

Task 1: Guided Activity, Course registration

```
import Foundation

var subjects = ["Swift", "UIKit", "SwiftUI"]

subjects.append("Networking")
subjects.insert("Testing", at: 1)
subjects.remove(at: subjects.firstIndex(of: "UIKit")!)

print("Sorted subjects:", subjects.sorted())

var registered: Set<String> = ["Anitha", "Megha", "Chitra"]

registered.insert("Anitha")
registered.insert("Brinda")

print("Registered students:", registered)
print("Number of students:", registered.count)

var marks: [String: Int] = [
    "Anitha": 85,
    "Megha": 92,
    "Chitra": 78
]
marks["Brinda"] = 88

marks["Anitha"] = 95

print("Esha's mark:", marks["Esha"] as Any)
for name in marks.keys.sorted() {
    print("\(name): \(marks[name]!)")
}

if let highest = marks.max(by: { $0.value < $1.value }) {
    print("Highest mark:", highest.value)
    print("Top student:", highest.key)
}
```

Output

```
Sorted subjects: ["Networking", "Swift", "SwiftUI", "Testing"]
Registered students: ["Chitra", "Brinda", "Anitha", "Megha"]
Number of students: 4
Esha's mark: nil
Anitha: 95
Brinda: 88
Chitra: 78
Megha: 92
Highest mark: 95
Top student: Anitha
```

Task 2: Practice Task, Word frequency counter

```
import Foundation

let sentence = "swift is fun and swift is fast"
let words = sentence.split(separator: " ").map { String($0) }
print("Total words:", words.count)
let uniqueWords = Set(words)
print("Unique words:", uniqueWords.sorted())
var frequency: [String: Int] = [:]
for word in words {
    frequency[word, default: 0] += 1
}
print("Word frequencies:")

for word in frequency.keys.sorted() {
    print("\(word): \(frequency[word]!)")
}
let rankedWords = frequency.sorted {
    if $0.value != $1.value {
        return $0.value > $1.value
    }
    return $0.key < $1.key
}
print("Ranked words:")

for (word, count) in rankedWords {
    print("\(word): \(count)")
}
```

Output

```
Total words: 7
Unique words: ["and", "fast", "fun", "is", "swift"]
Word frequencies:
and: 1
fast: 1
fun: 1
is: 2
swift: 2
Ranked words:
is: 2
swift: 2
and: 1
fast: 1
fun: 1
```

Task 3: Challenge Task, Set algebra and de-duplication

```
import Foundation

let swiftClub: Set<String> = ["Anitha", "Brinda", "Chitra", "Megha"]
let uiClub: Set<String> = ["Chitra", "Megha", "Esha"]
```

```swift
let both = swiftClub.intersection(uiClub)
print("Both clubs:", both.sorted())

let onlySwift = swiftClub.subtracting(uiClub)
print("Only Swift club:", onlySwift.sorted())

let everyone = swiftClub.union(uiClub)
print("Everyone:", everyone.sorted())

let exactlyOne = swiftClub.symmetricDifference(uiClub)
print("Exactly one club:", exactlyOne.sorted())
```

Output




```
Both clubs: ["Chitra", "Megha"]
Only Swift club: ["Anitha", "Brinda"]
Everyone: ["Anitha", "Brinda", "Chitra", "Esha", "Megha"]
Exactly one club: ["Anitha", "Brinda", "Esha"]
```